Step 1 — Verify GPU

In [1]:
```
!nvidia-smi
```

```
Wed Aug 26 04:56:45 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.82.07              Driver Version: 580.82.07      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   37C    P8              9W /   70W |       0MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

Step 2 — Install required libraries

In [2]:
```
!pip install -q diffusers transformers accelerate
```

Step 3 — Load the Stable Diffusion model

In [1]:
```
from diffusers import StableDiffusionPipeline
import torch

model_id = "runwayml/stable-diffusion-v1-5"

pipe = StableDiffusionPipeline.from_pretrained(
    model_id,
    torch_dtype=torch.float16
```

```
)
pipe = pipe.to("cuda")

print("Model loaded successfully!")
```

```
Flax classes are deprecated and will be removed in Diffusers v0.40.0. We recommen
d migrating to PyTorch classes or pinning your version of Diffusers.
Flax classes are deprecated and will be removed in Diffusers v0.40.0. We recommen
d migrating to PyTorch classes or pinning your version of Diffusers.
```

model_index.json: 100% 541/541 [00:00<00:00, 35.0kB/s]

Download complete: : 4.75GB, 18.0MB/s

Reconstruction complete: 100% 5.48GB / 5.48GB, 134MB/s

Fetching 15 files: 100% 15/15 [00:42<00:00, 5.82s/it]

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_
TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requ
ests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
```

Loading pipeline components...: 100% 7/7 [00:14<00:00, 2.41s/it]

Loading weights: 100% 396/396 [00:04<00:00, 82.74it/s]

Loading weights: 100% 196/196 [00:01<00:00, 134.38it/s]

```
Model loaded successfully!
```

Step 4 — Generate your first image

In [2]:
```
prompt = "a cat wearing a spacesuit, digital art, highly detailed"

image = pipe(prompt, num_inference_steps=30, guidance_scale=7.5).images[0]

image.save("generated_image.png")
image
```

100% 30/30 [00:05<00:00, 7.20it/s]

Out[2]:

In [3]:

```
prompt = "a cat wearing a spacesuit, digital art, highly detailed"

image_low_guidance = pipe(prompt, num_inference_steps=30, guidance_scale=3).imag
image_low_guidance.save("low_guidance.png")
image_low_guidance
```

100% 30/30 [00:04<00:00, 6.97it/s]

Out[3]: